Chapter 2 solutions

1

The consequent $p \to r$ is false only when $p = T, r = F$. Then $q = T$ makes $q \to r = F$, whereas $q = F$ makes $p \to q = F$. Thus the antecedent is false in both critical rows. All other rows have true consequent. No row has true antecedent and false consequent, so the formula is a tautology, the conditional form of hypothetical syllogism.

Let $H = (p \to q) \land (q \to r)$. All eight valuations, in F-to-T order, are:

p	q	r	$p \to q$	$q \to r$	H	$p \to r$	$H \to (p \to r)$
F	F	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	T	T	T	T	T	T	T
T	F	F	F	T	F	F	T
T	F	T	F	T	F	T	T
T	T	F	T	F	F	F	T
T	T	T	T	T	T	T	T

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The final column agrees with the critical-row argument.

2

From $\neg p$ and $p \lor q$, disjunctive syllogism gives q, since otherwise both premises could not hold. With $q \to r$, modus ponens gives r. The argument is valid.

3

$$\neg\exists x \forall y (P(x, y) \land x < y) \equiv \forall x \exists y \neg (P(x, y) \land x < y) \equiv \forall x \exists y (\neg P(x, y) \lor y \leq x).$$

Both variables range over one explicitly stated totally ordered domain, such as the reals, and P is binary. Replacing $\neg(x < y)$ by $y \leq x$ uses total order and fails for arbitrary partial orders. The original says no x satisfies both $P(x, y)$ and $x < y$ for every y.

4

Use contraposition. If integer n is not odd, it is even, say $n = 2k$. Then

$$3n + 2 = 6k + 2 = 2(3k + 1)$$

Chapter 2 solutions (continued)

is even, hence not odd. The contrapositive is true and equivalent to the desired implication.

5

Take $n = 9$. Its square 81 has remainder one modulo four, so the hypothesis holds. But nine has proper factor three and is not prime. It satisfies the hypothesis while falsifying the conclusion, as required.

6

The first is true. For each x, choose $y = -x$; the witness depends on x.

The second is false. A common y_0 would satisfy $x + y_0 = 0$ for every real x. Taking $x = 0$ forces $y_0 = 0$, and then $x = 1$ forces $1 = 0$. Quantifier order changes the common-witness requirement.

7

$$
\begin{aligned}
& \neg((p \vee q) \wedge \neg r) \vee q \\
\Leftrightarrow & (\neg(p \vee q) \vee \neg(\neg r)) \vee q && \text{De Morgan} \\
\Leftrightarrow & (\neg p \wedge \neg q) \vee r \vee q && \text{De Morgan and double negation} \\
\Leftrightarrow & (\neg p \wedge \neg q) \vee q \vee r && \text{commutativity} \\
\Leftrightarrow & (q \vee (\neg p \wedge \neg q)) \vee r && \text{commutativity} \\
\Leftrightarrow & [(q \vee \neg p) \wedge (q \vee \neg q)] \vee r && \text{distribution} \\
\Leftrightarrow & [(q \vee \neg p) \wedge T] \vee r && \text{excluded middle} \\
\Leftrightarrow & q \vee \neg p \vee r && \text{conjunction identity.}
\end{aligned}
$$

As a check, when $q = T$ the original is true. When $q = F$, it reduces to $\neg p \vee r$. Both cases agree with the result.

8

Use existential specification on $\exists x P(x)$, introducing a fresh witness name c and temporary $P(c)$. Universal specification gives $P(c) \to Q(c)$. Modus ponens yields $Q(c)$, and existential generalization yields $\exists x Q(x)$. Discharge the witness subproof, since its conclusion contains no free c.

Knowing the numerical identity of c is unnecessary. It must denote a domain element and obey the fresh-name conditions when the existential premise is used.

Chapter 2 solutions (continued)

9

No. Over integers let $P(x)$ mean even and $Q(x)$ mean odd. Every integer satisfies $P(x) \lor Q(x)$, so the antecedent is true. But $\forall x P(x)$ fails at one and $\forall x Q(x)$ fails at two. Both disjuncts of the consequent are false. This interpretation therefore makes the implication false.

For each x, at least one predicate holds, but no single predicate holds for all x. The choice may depend on x.

10

Assume for contradiction that $\sqrt{2} = a/b$, with integers a, b, $b \neq 0$, in reduced form $\gcd(a, b) = 1$. Squaring gives $a^2 = 2b^2$, so a^2 is even. The parity lemma, proved by contraposition in the lecture, implies a even. Put $a = 2k$. Then

$$(2k)^2 = 2b^2 \implies 4k^2 = 2b^2 \implies b^2 = 2k^2.$$

Thus b^2 is even, and a second use of the parity lemma makes b even. Both a, b have common divisor two, contradicting reduced form. The assumption fails, so $\sqrt{2}$ is irrational.

11

Over integers, let P be evenness and Q oddness. Zero witnesses P, and one witnesses Q. Both existential premises are true, but no integer satisfies both. The premises provide separate elements without evidence that they agree.

Existential elimination temporarily assumes $P(c)$ for fresh c and derives C in a subproof. Discharge that assumption only when c is free neither in C nor in other undischarged assumptions. A witness d for the second premise must also be fresh; one cannot impose $d = c$ without justification. The valid conclusion is $\exists x \exists y (P(x) \land Q(y))$, without identifying the variables.

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Chapter 2 solutions (continued)

12

p	q	r	φ
F	F	F	T
F	F	T	T
F	T	F	F
F	T	T	T
T	F	F	F
T	F	T	T
T	T	F	T
T	T	T	T

The canonical DNF is

$$\neg p \neg q \neg r \vee \neg p \neg q r \vee \neg p q r \vee p \neg q r \vee p q \neg r \vee p q r,$$

where adjacent literals abbreviate conjunction. Only 010 and 100 are false, giving CNF

$$(p \vee \neg q \vee r) \wedge (\neg p \vee q \vee r).$$

Each minterm is true only in its own row, and each maxterm false only in its own row. Both forms match φ row by row.

For $N(p, q) = \neg(p \wedge q)$,

$$p \vee q = N(N(p, p), N(q, q)).$$

NOT is $N(p, p)$, and AND is $N(N(p, q), N(p, q))$. Replace these three operations bottom-up in any DNF syntax tree. Structural induction preserves its truth function and yields a NAND-only formula. Constant T is $N(p, N(p, p))$, and $F = N(T, T)$, covering the always-true and always-false functions too.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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